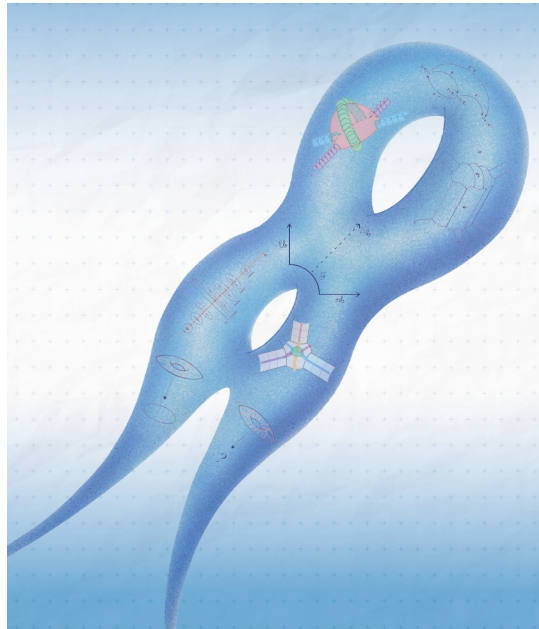

Pseudodifferential Techniques in Singular Analysis



31 August – 4 September 2026

*Mathematisches Institut
Georg-August-Universität Göttingen*



<https://pseudo-singular-analysis.github.io>

Scientific organizers:

Cipriana Anghel (Göttingen)
Clément Cren (Göttingen)
Erfan Rezaei (Göttingen)
Álvaro Sánchez Hernández (Hannover)
Rareș Stan (Bucharest)

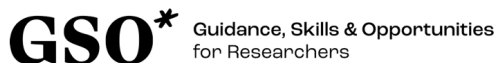
Administrative organizers:

Linda Haber (Göttingen)
Annalena Wendehorst (Göttingen)

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We gratefully acknowledge the hospitality and support of the [Mathematisches Institut Göttingen](#).

Many thanks to [Ioana Amza](#) for designing the conference poster and creating the accompanying artwork.



Klaus Tschira
Stiftung



Klaus Tschira
BOOST FUND



1. Schedule

Time	Monday 31 Aug	Tuesday 1 Sep	Wednesday 2 Sep	Thursday 3 Sep	Friday 4 Sep
08:00–09:00	Registration				
09:00–10:00	R. Melrose	I. Androulidakis	L. Lemoine	J. Rowlett	G. Jayasinghe
10:00–10:30	Coffee & Tea break				
10:30–11:30	F. Rochon	O. Mohsen	K. Bohlen	B. Ammann	T. Schick
11:45–12:45	S. Moroianu	V. F. Zenobi	J.-M. Lescure	C. Aldana	P. Piazza
12:45–14:30	Lunch break		Pizza lunch 12:45–13:45	Lunch break	
14:30–15:30	D. Grieser	V. Nistor	Lightning talks 13:45–15:00 Free afternoon	K. Fedosova	Departure
15:30–16:00	Coffee & Tea break			Coffee & Tea break	
16:00–17:00	J. O. Lye	M. Kohr		A. P. Contini	
19:30				Conference dinner	

Venue. All lectures take place in the **Maximum**, on the first floor of the Mathematisches Institut, Bunsenstraße 3–5, 37073 Göttingen.

Talks last 50 minutes, followed by 10 minutes for questions.

Pizza lunch on Wednesday, 12:45–13:45, followed by the **lightning talks** (Section 3): short contributed talks of 10 minutes.

Conference dinner: Thursday, 3 September, at *Ristorante Fellini*, Groner-Tor-Straße 28.

2. Titles and abstracts

Monday, 31 August

09:00–10:00

Richard Melrose

MIT

Simplicial compactification and pseudodifferential operators

I will start from the notion of a generalized product and the associated ‘small’ algebra of pseudodifferential operators and ‘large’ modules over it. After describing some of the properties and examples I will pass to the special case of the (in general conjectural) simplicial compactification of a Lie group. As time permits I will indicate possible implications of such a construction. This is based in part on joint work with Pierre Albin and with Chris Kottke and Michael Singer.

10:30–11:30

Frédéric Rochon*Université du Québec à Montréal****Construction of special non-Kähler metrics via the b -surgery calculus***

Apostolov, Lahdili and Lee recently launched a systematic study of 3-dimensional non-Kähler Ricci flat pluriclosed Hermitian manifolds (BHE) via their dimensional reduction to a special Kähler geometry in complex dimension 2. They notably provided the first non-Kähler examples of compact BHE 3-manifolds that are not locally isometric to a Samelson geometry. Building on these results, we will explain in this talk how to use the b -surgery pseudodifferential calculus of Mazzeo–Melrose to construct more examples by gluing ALE gravitational instantons to singular non-Kähler BHE manifolds. This is ongoing work with Apostolov, Lahdili and Lee.

11:45–12:45

Sergiu Moroianu*Universitatea din București & IMAR****Linking of invariant measures***

We define the linking number of two invariant measures for a flow on a closed 3-manifold, and prove its continuity with respect to the weak* topology on a subset of pairs of invariant measures. The definition relies on the asymptotic properties of the Green kernel on 1-forms in the direction of the flow. We deduce an existence result for Birkhoff sections for the Reeb flow on certain 3-dimensional contact homology spheres. This is joint work with Adrien Boulanger and Pierre Dehornoy.

14:30–15:30

Daniel Grieser*Universität Oldenburg****Cusps: geometry, microlocal analysis, spectral theory***

I will give a survey of some old and new results about cuspidal singularities, as well as their fibred generalizations and conformal relatives. I will start with the very basic question “What is a cuspidal metric?” and then discuss the microlocal structure of various operators associated to that geometry (inverses of elliptic operators, heat kernel, Dirichlet-to-Neumann operator), as well as some spectral theoretic results.

16:00–17:00

Jørgen Olsen Lye*University of Cologne****Winding and focusing for geodesics passing a thin cuspidal neck***

I will report on recent work with Daniel Grieser concerning the behaviour of geodesics on a family of manifolds which develop a conical or cuspidal singularity. Our main results are that geodesics will either start winding around the nascent singularity arbitrarily often, or they will approach the singularity from certain special directions. These results are obtained by a careful analysis of the geodesic flow as a Hamiltonian system, and some suitable rescaling procedure via blow-ups and a rescaled cotangent space.

Tuesday, 1 September

09:00–10:00

Iakovos Androulidakis*National and Kapodistrian University of Athens****Is the Helfer–Nourrigat conjecture geometric?***

We will survey a very well-known flow of ideas which go back to Riemann, in view of the recent proof of the Helffer–Nourrigat conjecture, wondering if this conjecture might be coming from geometry, after all.

10:30–11:30

Omar Mohsen

Université Paris Cité

On microlocal maximal hypoellipticity and applications

Given some vector fields satisfying Hormander’s condition, we will define a pseudo-differential calculus which contains the classical pseudo-differential calculus as well as the inhomogeneous pseudo-differential calculus (defined and studied by many people including among many others Folland, Stein, Rotschild, Beals, Taylor etc ...).

The main property of this calculus is that one can define a principal symbol which agrees with the classical principal symbol for classical pseudo-differential operators. Our main theorem is that invertibility of this symbol implies maximal hypoellipticity in the calculus. Using this result, we will obtain sufficient and necessary conditions for microlocal maximal hypoellipticity.

If time permits, we will also discuss other applications of this calculus to index theory and Egorov’s theorem.

11:45–12:45

Jean-Marie Lescure

Université Paris-Est Créteil

Deformation quantization on a class of Poisson manifolds, using groupoids and Toeplitz operators

For a prequantizable symplectic manifold M , V. Guillemin constructs an algebra of Toeplitz operators whose algebra of symbols is isomorphic to $C^\infty(M)[[t]]$, then providing a deformation quantization of the Poisson manifold M . This is later generalized by R. Melrose to any symplectic manifold. In this work with C. Cren and O. Mohsen, we have followed their approach to construct a deformation quantization for Poisson manifolds whose Poisson structure comes from a symplectic algebroid.

14:30–15:30

Victor Nistor

Université de Lorraine, Metz

Applications of operator algebras and operator theory to boundary value problems and layer potentials

One of the main and oldest approaches to boundary value problems is through layer potentials. The study of layer potentials has lead, more than one hundred years ago, to the development of operator theory (including compact and Fredholm operators). The layer potential operators associated to a suitable boundary value problem on a smooth, bounded domain are pseudodifferential operators and, in order to establish the solvability of the given boundary value problem, the basic theory of compact and Fredholm operators suffices. By contrast, on a non-smooth polyhedral domain, the relevant pseudodifferential and operator algebras have a more complicated structure and the basic theory of compact and Fredholm operators does not suffice anymore. In my talk, I will present some ideas from operator algebras that are useful in order to treat boundary value problems on domains with cylindrical ends (equivalently, on domains with conical points). Pseudodifferential operators similar to the ones considered by (Mazzeo, Melrose, Schrohe, Schulze and others) play an important role in establishing these connections. The main results of my talk are from joint works with Marius Mitrea and Mirela Kohr. (Applications of these results will be presented in Mirela Kohr’s talk.)

16:00–17:00

Mirela Kohr*Babeş-Bolyai University, Cluj-Napoca****Pseudodifferential operators, boundary value problems, and layer potentials on manifolds with cylindrical ends***

This talk is an interplay between the pseudodifferential operator theory, boundary value problems, and layer potentials, with applications to the Stokes type operator

$$\Xi := \Xi_{V,V_0} := \begin{pmatrix} L + V & \nabla \\ \nabla^* & -V_0 \end{pmatrix}$$

on a domain with straight cylindrical ends Ω . We use the method of layer potentials on a larger manifold with straight cylindrical ends M , $\Omega \subset M$. Under suitable positivity assumptions at infinity on V and V_0 , we prove that Ξ is Fredholm on $L^2(M; TM \oplus \mathbb{C})$. This allows us to define the single- and double-layer potential operators S and $\frac{1}{2}\mathbf{I} + K$. Under further positivity assumptions at infinity, we prove that these operators are also Fredholm. Under slightly stronger assumptions on V and V_0 (including non-negativity everywhere), we prove the invertibility of the operators Ξ , S , and $\frac{1}{2}\mathbf{I} + K$. The invertibility of these operators leads to well-posedness results for Stokes boundary value problems on domains with straight cylindrical ends. The proofs of these results require to develop an algebra of pseudodifferential operators that take into account the (translation invariant) structure of Ξ on manifolds with cylindrical ends. Joint work with Victor Nistor (Metz) and Wolfgang L. Wendland (Stuttgart).

Wednesday, 2 September

09:00–10:00

Lucas Lemoine*Université Paris-Est Créteil****Automorphisms of pseudodifferential operators on Lie groupoids***

Pseudodifferential operators form a filtered algebra that can be viewed as Fourier integral operators (FIOs) associated with the (anti)diagonal Lagrangian. Egorov's theorem states that this algebra is stable under conjugation by an FIO. Notably, as established by Duistermaat and Singer, the converse also holds: any automorphism of the filtered algebra of classical pseudodifferential operators can be realized as a conjugation by an invertible FIO. In this talk, I will present a generalization of this result to the setting of Lie groupoids, which provides a natural extension of classical analysis on smooth manifolds, encompassing some singular spaces. This involves a detailed examination of the symplectic structure of the cotangent groupoid, which is fundamental for constructing FIOs. Finally, I will discuss how this construction specializes to the setting of the b -calculus, describing its set of automorphisms.

10:30–11:30

Karsten Bohlen*Universität Regensburg****Nonlinear gradient flows on Lie groupoids***

I discuss semilinear and quasilinear heat equations on Lie groupoids arising as gradient flows of variational functionals. Among these, I consider a semilinear Schrödinger equation which is obtained by a Wick rotation in the time variable on Lie-structure groupoids, involving geometric Laplace-type operators acting along the longitudinal directions. Existence and uniqueness of solutions in suitable Sobolev and L^p Banach bundles, under critical smallness assumptions on the initial data, are shown. Another case that I intend to discuss is the heat flow on almost-injective Yang–Mills algebroid extensions. These extensions furnish a natural generalization of

the principal-bundle case, where they specialize to the standard Atiyah algebroid sequence. For the flow associated with these extensions, we describe a holonomy perturbation theory, or taming, in dimension four, which allows us to obtain a Fredholm operator after gauge fixing on manifolds with a boundary Lie structure. We prove local-in-time existence of solutions. This is based on a (Volterra) criterion for maximal regularity for realizations of tamed Laplace-type operators.

11:45–12:45

Vito Felice Zenobi

INdAM, Rome

Foliations, delocalized Chern characters and numeric ϱ -invariants

Let $(M, \mathcal{F})$ be a compact foliated manifold and let G be its monodromy groupoid. We can then define a long exact sequence of K -theory groups of C^* -algebras, called the adiabatic exact sequence, which is the receptacle of index theoretic invariants: the primary ones in the K -theory of the groupoid C^* -algebra of G and the secondary invariants in the K -theory group associated to its adiabatic deformation, the so-called ϱ -classes. Let us fix a complete transversal T of $(M, \mathcal{F})$ and the étale subgroupoid Γ of G obtained by restriction to T ; then for any dense and holomorphically closed subalgebra $\mathcal{A}(\Gamma)$ in $C_r^*(\Gamma)$, we are going to define a Chern type mapping from the adiabatic exact sequence to a long exact sequence of periodic homology groups, which in particular sends the ϱ -classes to the delocalized periodic homology group of $\mathcal{A}(\Gamma)$. Under suitable hypotheses about the foliation, through pairing with delocalized periodic cocycles, it is possible to extract numeric ϱ -invariants, associated to Dirac type operators, in order to obtain geometric and topological applications about $(M, \mathcal{F})$.

Thursday, 3 September

09:00–10:00

Julie Rowlett

Chalmers University and the University of Gothenburg

Extrema of determinants of Laplacians on tori

The search for extremal geometries is a central theme in several areas of mathematics. In this talk, we address the following question: among all n -dimensional orthogonal tori of unit volume, which one maximizes the determinant of the Laplacian? This determinant is obtained via the zeta regularization method, originally introduced by Ray and Singer to define an analytic analogue of the topological invariant, the Reidemeister torsion. While this determinant is connected to topology, geometry, analysis, and number theory, it is also important in physics as the zeta-function regularization technique can be used to regularize quadratic path integrals on a curved background space-time. We will find the maximizer and also observe what happens when the dimension increases and explore the physical interpretations.

10:30–11:30

Bernd Ammann

Universität Regensburg

The space of Dirac-minimal Riemannian metrics (joint work with Mattias Dahl)

Let M be a closed connected spin manifold of dimension m . For any Riemannian metric g on M we consider the Dirac operator D^g . In contrast to the Gauß-Bonnet operator $d + d^*$, the dimension of the kernel of D^g depends on g . The index $\alpha(M) \in KO_m(*)$ yields a lower bound on the dimension of the kernel of D^g . Riemannian metrics for which this lower bound is attained are called Dirac-minimal. We study the space $\mathcal{R}^{\min}(M)$ of all Dirac-minimal metrics. This space is always non-empty, open and dense in the space of all Riemannian metrics. Index theory provides tools to detect non-trivial homotopy and homology groups in $\mathcal{R}^{\min}(M)$. However, we

are interested in the interesting but hard question whether the space of Dirac-minimal metrics has topology not detected by index theory.

The main result of this talk states the absence of such topology in some simple cases: for dimensions $m = 2$ and $m = 4$, the space of Dirac-minimal metrics is connected. The reason is that the space of non-Dirac-minimal metrics is a stratified space of codimension at least 2. This is rather easy to show for the associated space of Fredholm operators, but requires some transversality statements for the space of metrics.

Full transversality fails in some cases for $m = 4$, e.g. on flat tori. But as we have partial transversality except for flat tori, we are still able to make the above conclusions.

11:45–12:45

Clara Aldana

Quasi-isospectrality of manifolds and Schrödinger operators.

In this talk, I will discuss quasi-isospectrality as a generalization of the classical notion of isospectrality. After briefly reviewing the isospectral problem and some results on isospectral Sturm–Liouville operators, I will introduce quasi-isospectral Schrödinger operators and discuss examples and constructions. I will then present recent results on quasi-isospectrality for Riemannian manifolds and Schrödinger operators. I will explain how quasi-isospectral closed manifolds of odd dimension are necessarily isospectral, while in even dimensions quasi-isospectrality has implications on the heat invariants. This is joint work with Camilo Perez (Universidad de los Andes).

14:30–15:30

Ksenia Fedosova

Universität Münster

Convolution identities for divisor functions

In this talk, we study convolution sums of divisor functions with (complex) indices. We show that a natural class of such convolution sums admits exact identities in terms of Fourier coefficients of Hecke cusp forms weighted by (not necessarily critical) values of their L -functions. The approach uses spectral methods and involves the Estermann zeta functions and the Petersson and Kuznetsov trace formula. This is joint work with Kim Klinger-Logan.

16:00–17:00

Alessandro Pietro Contini

Leibniz Universität Hannover

Singularities of the Klein-Gordon equation near a timelike boundary

The Klein-Gordon equation describes the motion of a spinless, massive particle on a spacetime with fixed background metric. As such, it is a fundamental building block for quantum field theory, and its theory is quite rich in the mathematical and physical literature. When the spacetime has a timelike boundary and a boundary condition is imposed, the analysis of the singularities that its solutions carry becomes more intricate. As these are fundamental to the scopes of QFT, including but not limited to the analysis of the Casimir force, much effort is currently being invested in constructing so-called Hadamard states. These are characterised, in the interior, by a condition on the wave-front set related to the causal structure. In this talk, based on joint work with Alexander Strohmaier, we will go over a general approach to the construction of these objects. We define a Dirichlet Hadamard state by the condition that the b -wave-front set of a certain Hilbert-space-valued distribution is contained in the compressed future causal cone. With the help of the b -calculus, but without using a general propagation of singularities theorem, we show that this condition is local-in-time. Finally, if time permits, we will show that in certain cases the states can be constructed with the help of spectral theory.

Friday, 4 September

09:00–10:00

Gayana Jayasinghe

Università di Roma

The L^2 equivariant Riemann–Roch theorem and its categorifications, on some stratified pseudomanifolds

The equivariant Riemann–Roch (ERR) theorem on complex manifolds is a powerful theorem, which admits extensions to settings with singularities and boundaries. Witten introduced holomorphic Morse inequalities that, in the case of Kähler Hamiltonian actions, generalize the ERR theorem, using his semiclassical approach to Morse theory in the presence of $N = 2$ SUSY, now called Witten deformation.

On certain complex spaces with singularities and boundaries, there are general index theorems that describe the Riemann–Roch defect (the difference between the Riemann–Roch number and the usual formula on the smooth open interior of the space) in terms of the eta invariant of a “boundary operator”, and its higher dimensional incarnations.

I will present an extension of the Witten instanton complex to stratified pseudomanifolds with Kähler wedge metrics with Kähler Hamiltonian actions. The space of spinors of the boundary operator corresponding to eta invariants at singularities (and at boundaries) admits a threefold direct sum decomposition, where two summands correspond to cohomology groups that are dual to each other, while the third does not contribute to the eta invariant due to “transverse/lifted supersymmetric cancellations”. These new symmetries bridge algebraic and analytic approaches to index theory on a large class of spaces. This is joint work with several collaborators.

10:30–11:30

Thomas Schick

Universität Göttingen

Abstract cone operators and applications to positive scalar curvature geometry

Conical singularities are among the fundamental types of singularities. There is a significant literature on appropriate calculi with notions of ellipticity and index theorems, which initiated a rather abstract functional analytic approach to these questions. In the talk, we describe a fully functional analytic setup for ellipticity and Fredholm theory. This expands on older work of Brüning and Seeley. The advantage is that it applies directly to low regularity operators. This allows us to use suspension constructions to prove rigidity results for positive scalar curvature metrics. This is joint work with Bernhard Hanke, Simone Cecchini and Lukas Schönlinner.

11:45–12:45

Paolo Piazza

Sapienza Università di Roma

The signature operator on Witt spaces: Gysin maps and applications

Gysin homomorphisms in K -theory and K -homology and their functoriality properties have played a major role in index theory. Using stable homotopy theory, Banagl has defined Gysin homomorphisms in K -homology for Witt pseudomanifolds and proved that they preserve the Sullivan–Siegel orientation class. Examples of Witt pseudomanifolds include complex projective varieties. In this talk I will explain an analytic approach to these results through the signature operator associated to a wedge metric. Crucial to this analytic approach is KK -theory and the edge pseudodifferential calculus of Mazzeo and Albin–Gell–Redman. This analytic approach will in fact allow for much more general results when compared to the topological approach. I will also discuss an application of these techniques to the formulation and proof of an Atiyah–Segal–Singer formula for the Goresky–MacPherson G -signature of a Witt pseudomanifold. The first part of the talk is joint work with Pierre Albin and Markus Banagl. The second part, on the G -signature

formula, is joint work with Markus Banagl and Eric Leichtnam. These articles, 2 + 1, are all available on the arXiv.

3. Contributed talks

Axel Gastaldi

Aix-Marseille Université

A Dirac-dual Dirac method in periodic cyclic homology

This talk proposes a homological analogue of the Connes–Kasparov isomorphism using an adaptation of the Dirac-dual Dirac method from bivariant K -theory to bivariant periodic cyclic homology. Our result establishes an isomorphism between the periodic cyclic homology of the smooth convolution algebras of a real Lie group G and of its maximal compact subgroup K , namely

$$HP_{\bullet}(\mathcal{C}_c^{\infty}(K) \otimes \mathcal{D}) \longrightarrow HP_{\bullet+\dim(G/K)}(\mathcal{C}_c^{\infty}(G) \otimes \mathcal{D}),$$

after stabilization by an inductive limit of Schatten ideals $\mathcal{D}$. Essential tools come from analytic estimations of the Dirac and dual Dirac elements as bivariant K -theory classes.

Pooja Joshi

Washington University in St. Louis

Lefschetz fixed point formula for subelliptic complexes on contact manifolds

In this talk, I will present a Lefschetz fixed point formula for subelliptic complexes on contact manifolds, extending the classical Atiyah–Bott theorem from the elliptic setting to the contact/sub-Riemannian setting. Using heat kernel methods, subelliptic Hodge theory, and the Heisenberg tangent groupoid, the formula expresses the Lefschetz number as a sum of contributions from the fixed points. Examples illustrate how the result connects global topological invariants with local analytic data determined by the contact structure.

Gustav Mårdby

Chalmers University of Technology

Heat trace asymptotics for alcoves

Alcoves form a very special class of polytopes for which the spectra of the Dirichlet and Neumann Laplacians can be computed explicitly. In this talk, I will explain how these explicit spectra can be used to study the short-time asymptotics of the associated heat traces, and how geometric quantities appear in the resulting expansions.

Florian Thiry

Institut de Mathématiques de Toulouse

Index theory for manifolds with corners

Depending on the time available, we will present some relative indices expressing the obstruction to full ellipticity for elliptic operators in geometries with corners.

Eva-Maria Hekkelman

Max Planck Institute for Mathematics, Bonn

Title to be announced

Paul Le Breton

Université Paris Cité

Title to be announced

4. Food and restaurants

The Mathematical Institute is at the southern border of the city centre, which means that there are a lot of restaurants to choose from. Here is a list of places we can recommend.

Quick lunch and bakeries

- **Zentralmensa** — the main campus cafeteria (Platz d. Göttinger Sieben 4), about a 20-minute walk from the Institute. Alternatively, take bus lines 91 or 92 from Bürgerstraße to Auditorium (about 14 minutes).
- **Bäckerei Küster** — Windausweg 8a or Weender Str. 106.
- **Göttinger Holzofenbäckerei** — Böttingerstraße 21.

German and European

- **Bullerjahn** — European cuisine, Markt 9.
- **Kartoffelhaus** — German cuisine, Goetheallee 8.
- **Zum Szültenbürger** — German cuisine, Prinzenstraße 7.

Italian

- **Tante Giulia** — Theaterstraße 25.
- **Ristorante Fellini** — Groner-Tor-Straße 28 (*conference dinner*).
- **Nudelhaus** — Rote Str. 13.

Asian

- **Sen Viet Cuisine Bistro** — Vietnamese, good vegetarian/vegan options, Rote Str. 18.
- **Asian Fusion Gamie** — sushi restaurant, Weender Str. 29.
- **Nam Anh** — Asian cuisine, Groner Str. 12.
- **Goa India** — Indian, good vegetarian/vegan options, Kurze-Geismar-Straße 43.
- **India Haus** — Indian, good vegetarian/vegan options, Kurze-Geismar-Straße 41.

Middle Eastern, African and South American

- **Palmyra** — Syrian cuisine, good vegetarian/vegan options, Düstere Str. 10.
- **Abessina** — Ethiopian restaurant, Ritterplan 2.
- **Empanadas Sabrosita** — South American cuisine, Karspüle 9.

Vegan and vegetarian

Several of the restaurants above also offer good vegetarian and vegan options. These are more specialized:

- **Gaia Garden** — Nikolaistraße 18.
- **Chay Vegan Kitchen** — Goethe-Allee 4A.

Cafés and sweets

- **Café Cortés** — a delicious cake shop.

5. The University and the city of Göttingen

- The University has a lot of **collections** which you can browse. In mathematics, we have in particular a beautiful **library** and a **collection of models**, calculating machines, integrators, etc. These can in part be viewed in the foyer in front of the lecture room.
- **Forum Wissen** — the new museum of knowledge in Göttingen.
- **Gänseliesel** — on the main market place in front of the old city hall. PhD students traditionally bring her flowers and give her a kiss following a successful defence.
- **Städtisches Museum Göttingen**.
- **Gauß’ grave** and the **Gauß Observatory**.
- **City Cemetery**, with the graves of David Hilbert, Max Born, Otto Hahn, Max Planck and many more.
- **Göttingen Forest**, including some easy and beautiful hikes (such as the Bismarckturm, the Göttinger Biergarten am Kehr, and the nearby *Tiergarten*).



The Gänseliesel fountain on the market place.